



HYPERSPECTRAL IMAGING FOR INTRAOPERATIVE BRAIN TUMOR DETECTION

FROM BENCHMARKS TO PREDICTION

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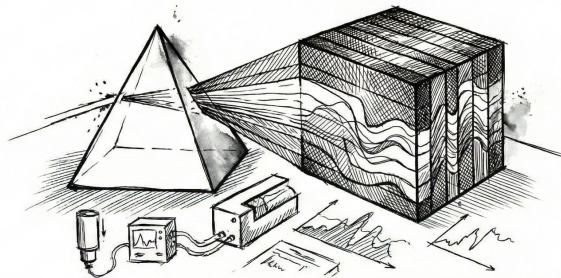




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The Clinical Problem

Challenges in Oncological Neurosurgery

- **Objective:** Achieve maximal brain tumor resection while preserving neurological functions (avoiding healthy tissue removal) and preventing hemorrhages (avoiding blood vessels).

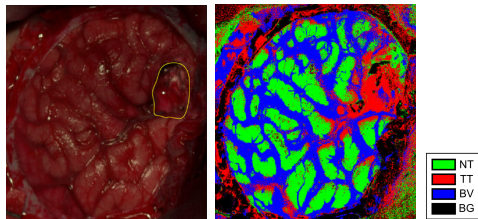


Figure: Example of brain imagery: (left) RGB image with tumor area circled; (right) brain segmentation into (**NT**= Normal Tissue, **TT**= Tumor Tissue, **BV**= Blood Vessels, **BG**= Background).



The Clinical Problem

Challenges in Oncological Neurosurgery

- **Objective:** Achieve maximal brain tumor resection while preserving neurological functions (avoiding healthy tissue removal) and preventing hemorrhages (avoiding blood vessels).
- **Challenges:** Blurred boundaries. Tumor cells intermingle with healthy tissues, making them visually indistinguishable to the naked eye.

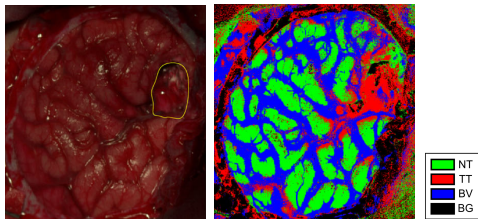


Figure: Example of brain imagery: (left) RGB image with tumor area circled; (right) brain segmentation into (**NT**= Normal Tissue, **TT**= Tumor Tissue, **BV**= Blood Vessels, **BG**= Background).



The Solution: Hyperspectral Imaging (HSI)

From RGB Vision to "Optical Biopsy"

The Hyperspectral Voxel

Unlike RGB sensors (3 bands), HSI captures hundreds of contiguous spectral bands. The base unit becomes the **voxel**, generating a 3D Hypercube (x, y, λ).

Spectral Signature:

- Functions as a biochemical tissue descriptor (oxygenation, hemoglobin).
- Enables *non-contact* classification.
- Leverages tumor metabolic alterations to map tumor margins.

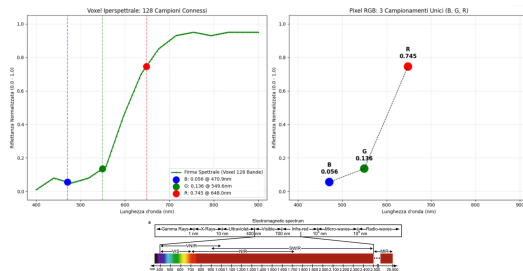




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The InVivoBench Dataset

Characteristics and Limitations

Context: *In-vivo* data acquired during real neurosurgical procedures (HELICoiD Project). 61 images from 34 patients.

The Class Imbalance Problem

The dataset exhibits a severe class imbalance. Tumor Tissue (TT) represents a minimal fraction of the total volume.

Class	Voxel Count	Percentage
Unlabeled	11,805,660	92.89%
Background (BG)	445,540	3.51%
Normal Tissue (NT)	308,328	2.43%
Blood Vessel (BV)	112,407	0.88%
Tumor Tissue (TT)	37,399	0.29%



Validation and Data Leakage Prevention

Custom Patient-Level 5-Fold Cross-Validation

- **Risk:** Random voxel-level splitting causes *Data Leakage*, as voxels from the same patient are highly correlated.
- **The Solution (Patient-Level Split):**
 - Partitioning based on unique Patient IDs.
 - Internal division: **60% Training / 20% Validation / 20% Test** for each fold.
- **Clinical Significance:** Reflects the ideal surgical scenario, where the system operates on a *new* subject never seen during training.

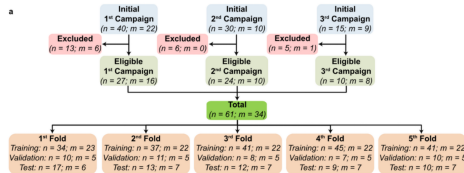


Figure: Patient-based rotation scheme. The Test set always isolates a previously unseen patient.



Performance Analysis

The Sensitivity Issue

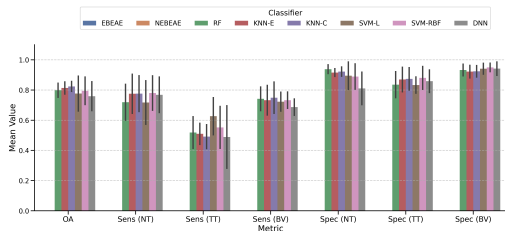


Figure: Sensitivity and Specificity details per class

- **Optimal Specificity:** All models demonstrate a high capacity to reject false positives (global Specificity $> 80\text{-}90\%$).
- **Sensitivity** in the Tumor Tissue (TT) class remains the structural weakness for all classifiers.
- **False Negatives:** Models struggle to correctly identify *all* tumor voxels, often misclassifying them as vessels (BV) or healthy tissue (NT).



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Diagnostic Error Analysis

The Spectral Ambiguity Phenomenon

- Why do classifiers fail on tumor tissue? Analysis of Tumor False Negatives reveals that misclassified voxels do **not** exhibit a noisy or degraded signal.
- Spectral signatures of erroneous voxels perfectly mimic the class they are confused with.
- *Example:* A hypervascularized tumor is spectrally almost indistinguishable from a blood vessel (BV).

Analisi Spettrale Errori Falsi Negativi (FN) per SVM-L

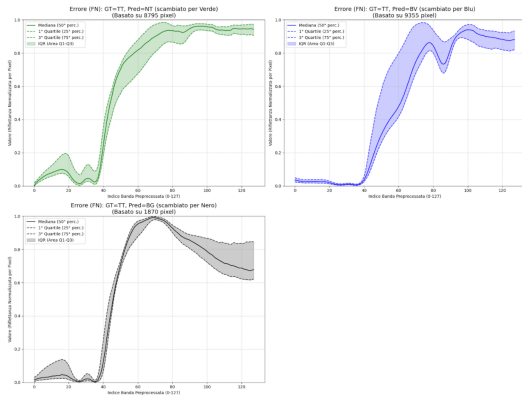


Figure: False Negative Spectra (SVM)

Diagnostic Error Analysis

Visual Comparison: Spectral Ambiguity

Analisi Statistica delle Firme Spettrali (Mediana e IQR)

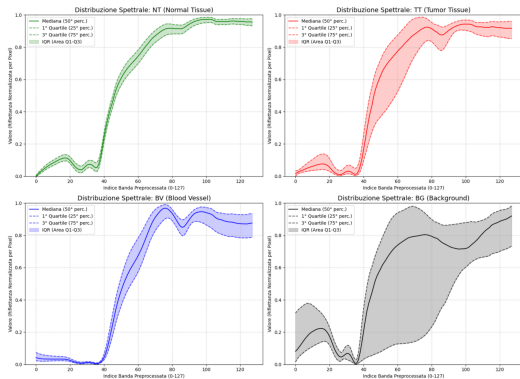


Figure: Real Spectral Signatures (Ground Truth)

Analisi Spettrale Errori Falsi Negativi (FN) per SVM-L

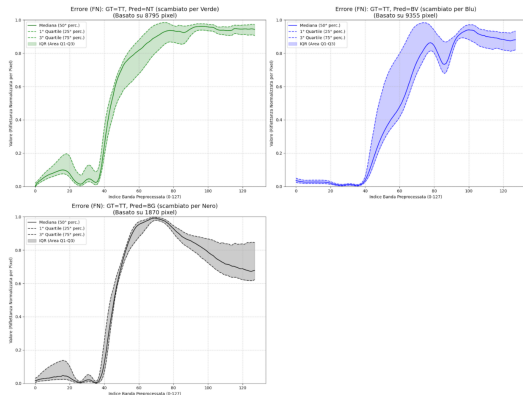


Figure: False Negative Signatures (Model Errors)



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Introduction to Ensemble Learning

From single predictions to collective intelligence

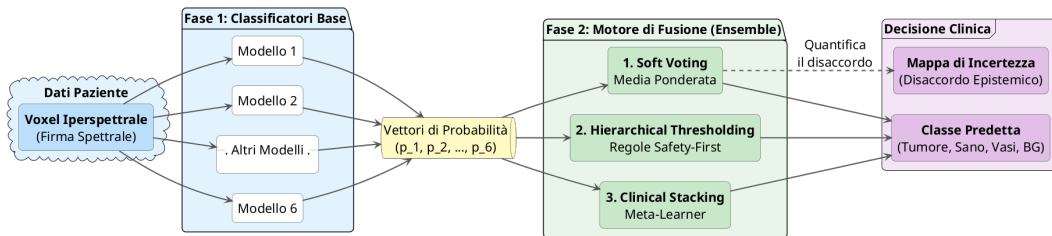


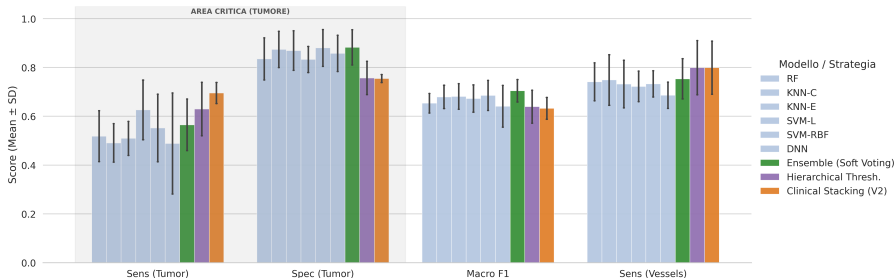
Figure: Probabilities extracted from 6 base classifiers serve as input for a fusion engine (Ensemble), tasked with generating the clinical decision and quantifying uncertainty.



Ensemble Results

Accuracy vs. Safety Trade-off

Analisi Dettagliata: Modelli Singoli vs Strategie Ensemble



- **Soft Voting:** Achieves the best global balance (Macro F1: 0.71).
- **Clinical Stacking / Hierarchical:** Sacrifice specificity (more false positives) to **maximize sensitivity** for Tumor and Blood Vessels.

The Role of Epistemic Uncertainty

From prediction to reliability assessment

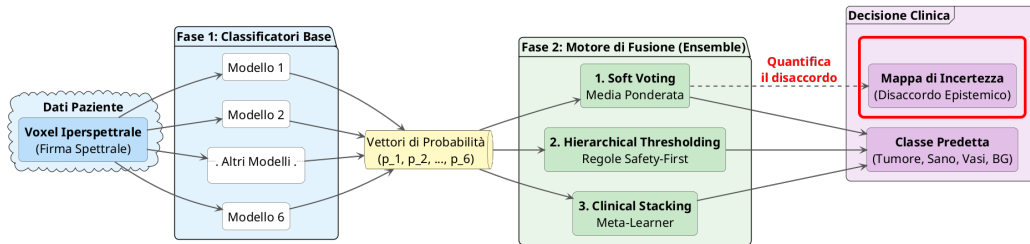


Figure: By leveraging divergence between base models, the Ensemble provides more than just a rigid classification; it extracts an **uncertainty map** to guide critical surgical decisions.



Epistemic Uncertainty Analysis

Quantifying Prediction Reliability

The probabilistic ensemble allows for decomposing uncertainty to isolate **model disagreement**:

- **Total** (U_{tot}): Ensemble Entropy.

$$U_{tot} = - \sum \bar{p}_c \log_K(\bar{p}_c)$$

- **Aleatoric** (U_{ale}): Data noise.

$$U_{ale} = \frac{1}{M} \sum H(P_m)$$

- **Epistemic** (U_{epi}): Disagreement.

$$U_{epi} = U_{tot} - U_{ale}; U_{epi} \in [-1, 1]$$

Clinical Impact (r_{pb}): The global statistical correlation is **low** (≈ 0.20), however...

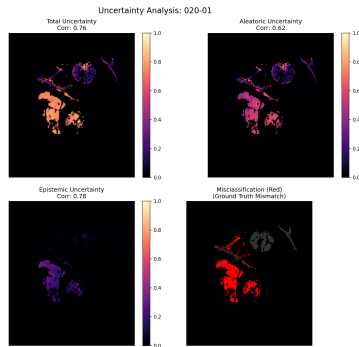


Figure: Uncertainty Maps (Patient 020-01)



Operational Validation

Risk-Rejection Curves

What happens if the system abstains from classifying the most uncertain voxels, delegating them to the surgeon?

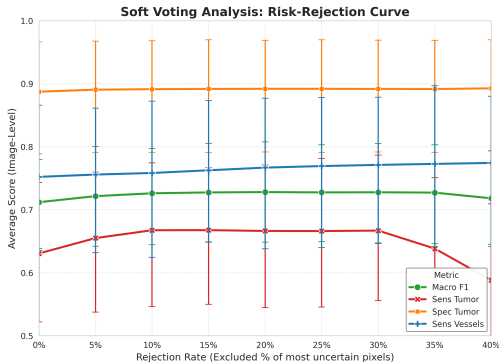


Figure: Risk-Rejection (Soft Voting)

- By rejecting the top 10% **voxels** with the highest Epistemic Uncertainty, **tumor sensitivity (red line) increases by 4 points**, rising from 63% to 67%.
- The Soft Voting Ensemble thus approaches the tumor sensitivity levels of the **Clinical Stacker**, while maintaining higher tumor specificity ($\sim 90\%$).



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Conclusions and Future Work

5 Conclusions

Summary of Results

HSI is a promising technology; however, voxel-wise classification suffers from intrinsic dataset ambiguities.

Ensemble architectures, combined with uncertainty analysis, offer a robust and clinically interpretable solution for surgeons.

Future Directions:

- Implement more robust labeling for HSI cubes, validating labels with intraoperative MRI.
- Develop *Human-in-the-loop* interfaces to visualize risk maps during surgery.



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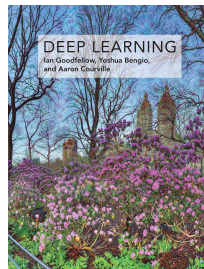
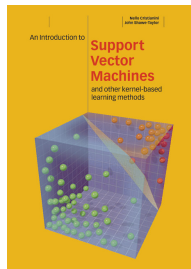
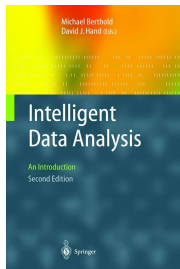
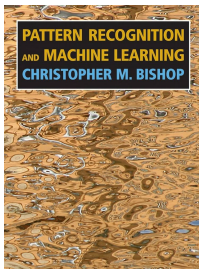
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Epistemic Uncertainty and Statistical Validation

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**Aleatoric and epistemic uncertainty in machine learning:
an introduction to concepts and methods**

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Abstract
The notion of uncertainty is of major importance in machine learning and constitutes a key element of machine learning methodologies. In line with the statistical tradition, uncertainty has long been perceived as almost synonymous with standard probability and probabilistic predictions. Yet, due to the steadily increasing relevance of machine learning for practical applications and related issues such as safety requirements, new problems and challenges have recently been identified by machine learning scholars, and these problems may call for new methodological developments. In particular, this includes the importance of distinguishing between (at least) two different types of uncertainty, often referred to as aleatoric and epistemic. In this paper, we provide an introduction to the state of uncertainty in machine learning as well as an overview of attempts so far at handling uncertainty in general and formalizing this distinction in particular.

Keywords Uncertainty · Probabilistic · Epistemic uncertainty · Bayesian epistemic learning · Bayesian inference · Calibration · Ensemble · Gaussian processes · Deep neural networks · Likelihood-based methods · Credit sets and classifiers · Conformal prediction · Set-valued prediction · Generative models

1 Introduction

Machine learning is essentially concerned with extracting models from data, often (though not exclusively) using them for the purpose of prediction. As such, it is inseparably connected with uncertainty. Indeed, learning in the sense of generalizing beyond the data seen

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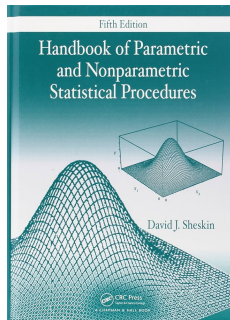
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Thank you for your attention!